AP CALCULUS AB

Unit 1 • Worksheet 1-3A • Answer key

Worksheet 1-3A • Limits to Infinity and Horizontal Asymptotes

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Which describes the graph of $f(x) = \dfrac{x+1}{x^2-4x-5}$?

(A) A hole at $x = -1$ and a vertical asymptote at $x = 5$

(B) Vertical asymptotes at $x = -1$ and $x = 5$

(C) A hole at $x = 5$ and a vertical asymptote at $x = -1$

(D) No hole and no vertical asymptote

Solution

Answer: (A). $\frac{x+1}{(x-5)(x+1)}$: $x+1$ cancels (hole), $x-5$ stays (asymptote).

Question 2

NO CALCULATOR

What is the y-coordinate of the hole in the graph of $f(x) = \dfrac{x+1}{x^2-4x-5}$?

(A) 0

(B) $-\frac{1}{6}$

(C) $\frac{1}{6}$

(D) $-\frac{1}{4}$

Solution

Answer: (B). After cancelling, $\frac{1}{x-5}$ at $x = -1$: $-\frac{1}{6}$.

Question 3

NO CALCULATOR

For the same f, what is $\lim_{x \to 5^+} f(x)$?

(A) $-\infty$

(B) 0

(C) $+\infty$

(D) $\frac{1}{6}$

Solution

Answer: (C). $\frac{1}{x-5}$ with $x-5$ small and positive: $+\infty$.

Question 4

NO CALCULATOR

What is $\lim_{x\to\infty} \dfrac{2x^2-3}{5x^2+x}$?

(A) $\frac{2}{5}$
(B) 0
(C) -3
(D) It does not exist.

Solution
Answer: (A). Same degree: $\frac{2}{5}$.

Question 5

NO CALCULATOR

What is $\lim_{x\to\infty} \dfrac{8x-1}{2x^3+5}$?

(A) 4
(B) 8
(C) It does not exist.
(D) 0

Solution
Answer: (D). The bigger degree is on the bottom: 0.

Question 6

NO CALCULATOR

What is $\lim_{x\to-\infty} \dfrac{x^3+1}{4x-x^2}$?

(A) -1
(B) It does not exist (the function grows without bound).
(C) 0
(D) $\frac{1}{4}$

Solution
Answer: (B). The bigger degree is on top, so there is no horizontal asymptote ($\frac{x^3}{-x^2} = -x \to +\infty$).

Question 7

NO CALCULATOR

Which are all the asymptotes of $y = \dfrac{3x^2+1}{x^2-9}$?

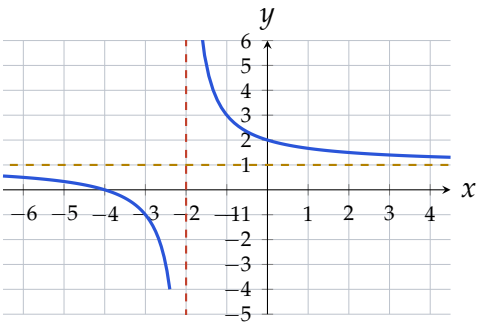
(A) $x = 3$ and $y = 3$ only
(B) $x = -3$, $x = 3$ and $y = 0$
(C) $x = -3$, $x = 3$ and $y = 3$
(D) $y = 3$ only

Solution
Answer: (C). Nothing cancels: $x = \pm 3$ are vertical asymptotes. Same degree: HA $y = \frac{3}{1} = 3$.

Question 8

NO CALCULATOR

The graph of g is shown, with its asymptotes. What is $\lim_{x \to -2^-} g(x)$?



(A) 1
(B) $+\infty$
(C) 0
(D) $-\infty$

Solution
Answer: (D). Just to the left of $x = -2$, the graph goes down without bound.

Question 9 (Free response)

NO CALCULATOR

Let $k(x) = \dfrac{2x^2 - 2}{x^2 + 2x - 3}$.

(a) Factor k. Find the vertical asymptote and the hole (both coordinates).
(b) Find $\lim_{x \to \infty} k(x)$ and $\lim_{x \to -\infty} k(x)$. What is the horizontal asymptote?
(c) Find $\lim_{x \to -3^+} k(x)$.

Solution
(a) $k = \frac{2(x-1)(x+1)}{(x-1)(x+3)}$. VA $x = -3$; hole at $x = 1$, height $\frac{2(2)}{4} = 1$: $(1, 1)$.
(b) Same degree: both limits are 2; HA $y = 2$.
(c) $\frac{2(x+1)}{x+3}$: numerator near -4, denominator small and positive: $-\infty$.